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Промышленное развертывание серверных приложений

Домашняя работа 4 – helm

На 6: dev-режим

Рендер манифестов: helm template kc-dev . -f ./values-dev.yaml --set admin.password=x

```
PS C:\Users\prado\IdeaProjects\MineCoin\helm> helm template kc-dev . -f ./values-dev.yaml --set admin.password=x
---
# Source: keycloak/templates/secret.yaml
apiVersion: v1
kind: Secret
metadata:
  name: kc-dev-keycloak-admin
type: Opaque
stringData:
  username: "admin"
  password: "x"
---
# Source: keycloak/templates/deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: kc-dev-keycloak
  labels:
    app: kc-dev-keycloak
spec:
  replicas: 1
  selector:
    matchLabels:
      app: kc-dev-keycloak
  template:
    metadata:
      labels:
        app: kc-dev-keycloak
    spec:
      containers:
        - name: keycloak
          image: "quay.io/keycloak/keycloak:26.4.0"
          imagePullPolicy: IfNotPresent
          args: ["start-dev"]
          env:
            - name: KC_BOOTSTRAP_ADMIN_USERNAME
              valueFrom:
                secretKeyRef:
                  name: kc-dev-keycloak-admin
                  key: username
            - name: KC_BOOTSTRAP_ADMIN_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: kc-dev-keycloak-admin
                  key: password
            - name: KC_HEALTH_ENABLED
              value: "true"
            - name: KC_METRICS_ENABLED
              value: "true"
          ports:
            - name: http
              containerPort: 8080
            - name: https
              containerPort: 8443
            - name: management
              containerPort: 9000
          startupProbe:
            httpGet:
              path: /health/started
              port: management
              scheme: HTTP
            periodSeconds: 5
            failureThreshold: 60
          livenessProbe:
            httpGet:
              path: /health/live
              port: management
              scheme: HTTP
            periodSeconds: 10
            failureThreshold: 3
          readinessProbe:
            httpGet:
              path: /health/ready
              port: management
              scheme: HTTP
            periodSeconds: 5
            failureThreshold: 3
          resources:
            limits:
              memory: 1Gi
            requests:
              cpu: 250m
              memory: 512Mi
---
# Source: keycloak/templates/tests/test-health.yaml
apiVersion: v1
kind: Pod
metadata:
  name: kc-dev-test-health
  annotations:
    "helm.sh/hook": test
    "helm.sh/hook-delete-policy": before-hook-creation
spec:
  restartPolicy: Never
  containers:
    - name: check-health
      image: curlimages/curl
      command:
        - sh
        - -c
        - curl -fsk http://kc-dev-keycloak:9000/health/ready
PS C:\Users\prado\IdeaProjects\MineCoin\helm>
```

Лог Keycloak, Profile dev activated: kubectl logs deployment/kc-dev-keycloak -n k8s-lab

```
PS C:\Users\prado\IdeaProjects\MineCoin\helm> kubectl logs deployment/kc-dev-keycloak -n k8s-lab
Updating the configuration and installing your custom providers, if any. Please wait.
2026-10-15 17:39:41,685 INFO  [io.quarkus.deployment.QuarkusAugmentor] (main) Quarkus augmentation completed in 5302ms
Running the server in development mode. DO NOT use this configuration in production.
2026-10-15 17:39:45,025 INFO  [org.hibernate.orm.jdbc.batch] (JPA Startup Thread) HHH100501: Automatic JDBC statement ba
tching enabled (maximum batch size 32)
2026-10-15 17:39:45,976 INFO  [org.keycloak.quarkus.runtime.storage.database.liquibase.QuarkusJpaUpdaterProvider] (main)
 Initializing database schema. Using changelog META-INF/jpa-changelog-master.xml
2026-10-15 17:39:52,100 INFO  [org.infinispan.CONTAINER] (main) Virtual threads support enabled
2026-10-15 17:39:52,208 INFO  [org.infinispan.CONTAINER] (main) ISPN000556: Starting user marshaller 'org.infinispan.com
mons.marshall.ImmutableProtoStreamMarshaller'
2026-10-15 17:39:52,590 INFO  [org.keycloak.connections.infinispan.DefaultInfinispanConnectionProviderFactory] (main) No
de name: node_860619, Site name: null
2026-10-15 17:39:52,825 INFO  [org.keycloak.services] (main) KC-SERVICES0050: Initializing master realm
2026-10-15 17:39:54,570 INFO  [org.keycloak.services] (main) KC-SERVICES0077: Created temporary admin user with username
 admin
2026-10-15 17:39:54,718 INFO  [io.quarkus] (main) Keycloak 26.4.0 on JVM (powered by Quarkus 3.27.0) started in 12.937s.
 Listening on: http://0.0.0.0:8080. Management interface listening on http://0.0.0.0:9000.
2026-10-15 17:39:54,718 INFO  [io.quarkus] (main) Profile dev activated.
2026-10-15 17:39:54,719 INFO  [io.quarkus] (main) Installed features: [agroal, cdi, hibernate-orm, jdbc-h2, keycloak, mi
crometer, narayana-jta, opentelemetry, reactive-routes, rest, rest-jackson, smallrye-context-propagation, smallrye-healt
h, vertx]
PS C:\Users\prado\IdeaProjects\MineCoin\helm>
```

Проверка готовности: helm test kc-dev -n k8s-lab

```
PS C:\Users\prado\IdeaProjects\MineCoin> helm test kc-dev -n k8s-lab
NAME: kc-dev
LAST DEPLOYED: Thu Oct 15 20:50:50 2026
NAMESPACE: k8s-lab
STATUS: deployed
REVISION: 2
DESCRIPTION: Upgrade complete
TEST SUITE:     kc-dev-test-health
Last Started:   Thu Oct 15 20:52:24 2026
Last Completed: Thu Oct 15 20:52:45 2026
Phase:          Succeeded
```

На 8: продовый режим

Установка: helm install kc-prod helm -n k8s-lab -f helm/values-prod.yaml --set ...

```
PS C:\Users\prado\IdeaProjects\MineCoin> helm install kc-prod helm -n k8s-lab -f helm/values-prod.yaml --set admin.passw
ord=ProdPassword1 --set postgresql.auth.password=KcDbPassword1 --set postgresql.auth.postgresPassword=PgAdminPassword1
NAME: kc-prod
LAST DEPLOYED: Thu Oct 15 21:01:20 2026
NAMESPACE: k8s-lab
STATUS: deployed
REVISION: 1
DESCRIPTION: Install complete
NOTES:
Keycloak (prod) ▼▼▼▼▼▼▼▼▼▼.
▼▼▼▼▼▼:
  kubectl port-forward svc/kc-prod-keycloak 8080:8080 -n k8s-lab
  http://localhost:8080
PS C:\Users\prado\IdeaProjects\MineCoin>
```

Запуск подов, Keycloak ждёт PostgreSQL в Init:0/1: kubectl get pods -n k8s-lab -w

```
PS C:\Users\prado\IdeaProjects\MineCoin> kubectl get pods -n k8s-lab -w
NAME                                 READY   STATUS              RESTARTS   AGE
kc-dev-test-health                   0/1     Completed           0          8m24s
kc-prod-keycloak-54ddbd947f-8nf66    0/1     Init:0/1            0          16s
kc-prod-keycloak-54ddbd947f-z2qnx    0/1     Init:0/1            0          16s
kc-prod-postgresql-0                 0/1     ContainerCreating   0          16s
kc-prod-keycloak-54ddbd947f-8nf66    0/1     Init:0/1            0          41s
kc-prod-keycloak-54ddbd947f-z2qnx    0/1     Init:0/1            0          45s
kc-prod-postgresql-0                 0/1     Running             0          72s
kc-prod-postgresql-0                 1/1     Running             0          83s
kc-prod-postgresql-0                 1/1     Running             0          83s
kc-prod-keycloak-54ddbd947f-z2qnx    0/1     PodInitializing     0          84s
kc-prod-keycloak-54ddbd947f-8nf66    0/1     PodInitializing     0          84s
kc-prod-keycloak-54ddbd947f-8nf66    0/1     Running             0          84s
kc-prod-keycloak-54ddbd947f-8nf66    0/1     Running             0          85s
```

Лог Keycloak, Profile prod activated и jdbc-postgresql: kubectl logs deployment/kc-prod-keycloak -n k8s-lab

```
PS C:\Users\prado\IdeaProjects\MineCoin> kubectl logs deployment/kc-prod-keycloak -n k8s-lab
Found 2 pods, using pod/kc-prod-keycloak-54ddbd947f-8nf66
Defaulted container "keycloak" out of: keycloak, wait-for-postgres (init)
Changes detected in configuration. Updating the server image.
Updating the configuration and installing your custom providers, if any. Please wait.
2026-10-15 18:02:50,872 INFO  [io.quarkus.deployment.QuarkusAugmentor] (main) Quarkus augmentation complet
ed in 5378ms
Server configuration updated and persisted. Run the following command to review the configuration:

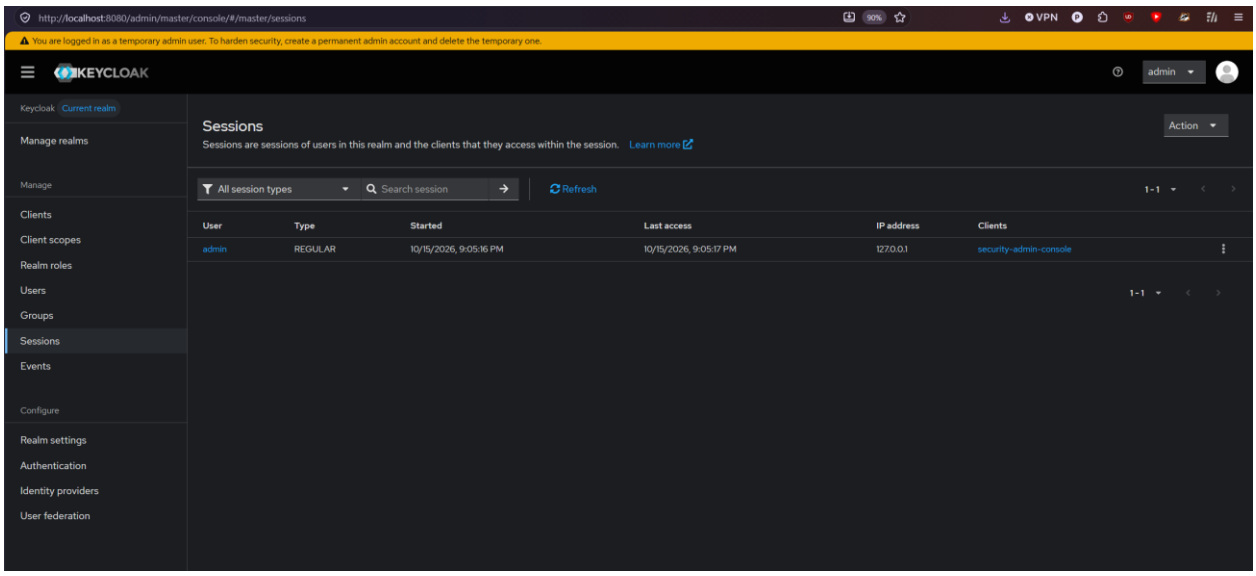
        kc.sh show-config

Next time you run the server, just run:

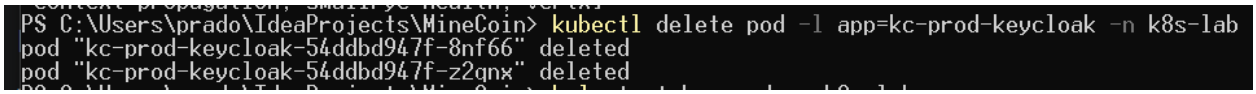
        kc.sh start --optimized

2026-10-15 18:02:54,751 INFO  [org.hibernate.orm.jdbc.batch] (JPA Startup Thread) HHH100501: Automatic JDB
C statement batching enabled (maximum batch size 32)
2026-10-15 18:02:55,957 INFO  [org.keycloak.quarkus.runtime.storage.database.liquibase.QuarkusJpaUpdaterPr
ovider] (main) Initializing database schema. Using changelog META-INF/jpa-changelog-master.xml
2026-10-15 18:03:04,326 INFO  [org.keycloak.spi.infinispan.impl.embedded.JGroupsConfigurator] (main) JGrou
ps JDBC_PING discovery enabled.
2026-10-15 18:03:04,594 INFO  [org.keycloak.spi.infinispan.impl.embedded.JGroupsConfigurator] (main) JGrou
ps Encryption enabled (mTLS).
2026-10-15 18:03:04,717 INFO  [org.infinispan.CONTAINER] (main) Virtual threads support enabled
2026-10-15 18:03:04,782 INFO  [org.keycloak.jgroups.certificates.CertificateReloadManager] (main) Starting
 JGroups certificate reload manager
2026-10-15 18:03:04,902 INFO  [org.infinispan.CONTAINER] (main) ISPN000556: Starting user marshaller 'org.
infinispan.commons.marshall.ImmutableProtoStreamMarshaller'
2026-10-15 18:03:05,055 INFO  [org.infinispan.CLUSTER] (main) ISPN000078: Starting JGroups channel 'ISPN'
with stack 'jdbc-ping'
2026-10-15 18:03:05,056 INFO  [org.jgroups.JChannel] (main) local_addr: 00000000-0000-0000-0000-0000000000
01, name: kc-prod-keycloak-54ddbd947f-8nf66-39158
2026-10-15 18:03:05,061 INFO  [org.jgroups.protocols.FD_SOCK2] (main) server listening on *:57800
2026-10-15 18:03:05,085 INFO  [org.infinispan.CLUSTER] (main) ISPN000094: Received new cluster view for ch
annel ISPN: [kc-prod-keycloak-54ddbd947f-8nf66-39158|0] (1) [kc-prod-keycloak-54ddbd947f-8nf66-39158]
2026-10-15 18:03:05,088 INFO  [org.keycloak.jgroups.certificates.CertificateReloadManager] (main) Reloadin
g JGroups Certificate
2026-10-15 18:03:05,122 INFO  [org.infinispan.CLUSTER] (main) ISPN000079: Channel 'ISPN' local address is
'kc-prod-keycloak-54ddbd947f-8nf66-39158', physical addresses are '[10.244.2.4:7800]'
2026-10-15 18:03:05,486 INFO  [org.keycloak.connections.infinispan.DefaultInfinispanConnectionProviderFact
ory] (main) Node name: kc-prod-keycloak-54ddbd947f-8nf66-39158, Site name: null
2026-10-15 18:03:05,636 INFO  [org.keycloak.services] (main) KC-SERVICES0050: Initializing master realm
2026-10-15 18:03:07,070 INFO  [org.keycloak.services] (main) KC-SERVICES0077: Created temporary admin user
 with username admin
2026-10-15 18:03:07,187 INFO  [io.quarkus] (main) Keycloak 26.4.0 on JVM (powered by Quarkus 3.27.0) start
ed in 16.210s. Listening on: http://0.0.0.0:8080. Management interface listening on http://0.0.0.0:9000.
2026-10-15 18:03:07,188 INFO  [io.quarkus] (main) Profile prod activated.
2026-10-15 18:03:07,188 INFO  [io.quarkus] (main) Installed features: [agroal, cdi, hibernate-orm, jdbc-po
stgresql, keycloak, micrometer, narayana-jta, opentelemetry, reactive-routes, rest, rest-jackson, smallrye
-context-propagation, smallrye-health, vertx]
PS C:\Users\prado\IdeaProjects\MineCoin>
```

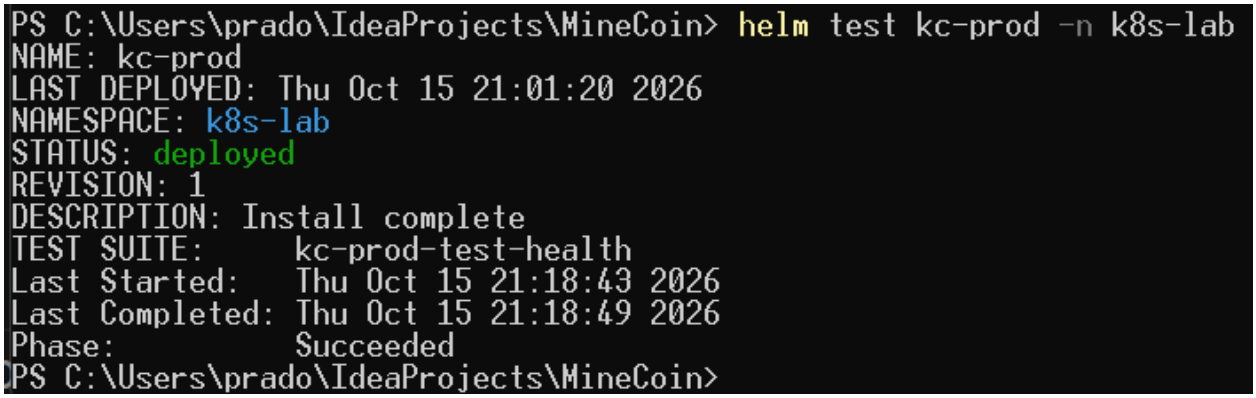
Админка через kubectl port-forward svc/kc-prod-keycloak 8080:8080 -n k8s-lab, сессия начата в 21:05:16



Удаление подов Keycloak: kubectl delete pod -l app=kc-prod-keycloak -n k8s-lab



Проверка готовности: helm test kc-prod -n k8s-lab



На 10: самоподписанный сертификат

Выпуск сертификата: openssl req -x509 -newkey rsa:2048 -nodes -days 365 -keyout tls.key -out tls.crt -subj "/CN=localhost" -addext "subjectAltName=DNS:localhost,IP:127.0.0.1"

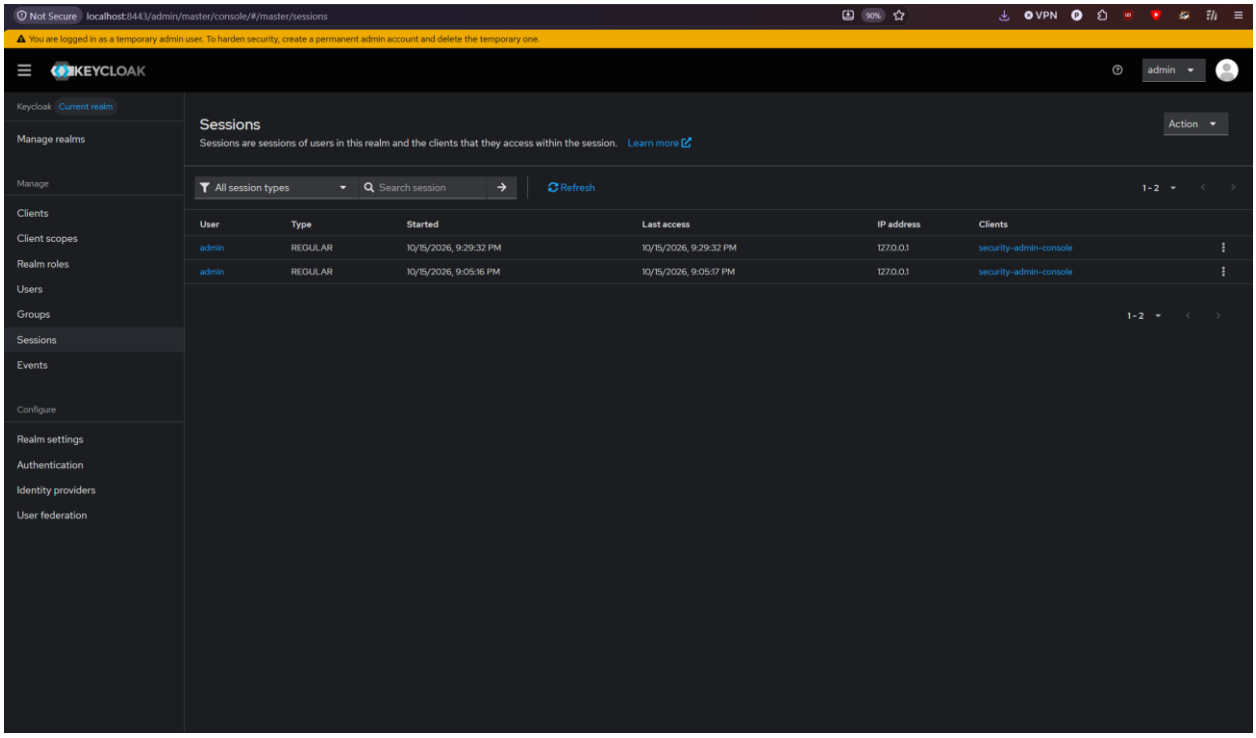


После этого сертификат загружен в кластер и включён HTTPS:

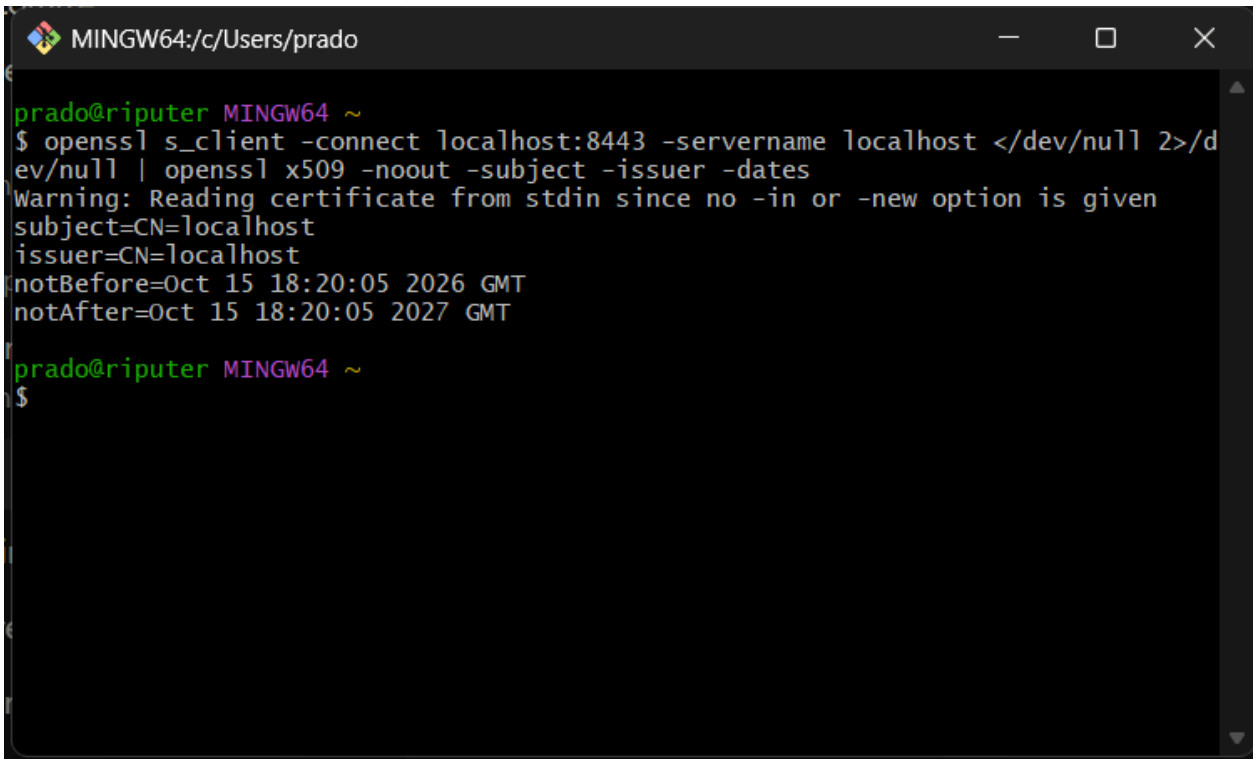
kubectl create secret tls keycloak-tls --cert=helm/tls.crt --key=helm/tls.key -n k8s-lab

helm upgrade kc-prod helm -n k8s-lab -f helm/values-prod.yaml -f helm/values-prod-tls.yaml --set ...

Админка через kubectl port-forward svc/kc-prod-keycloak 8443:8443 -n k8s-lab: браузер показывает Not Secure, а сессия из 21:05:16 сохранилась после замены всех подов, потому что данные лежат в PostgreSQL



Сертификат, который отдаёт Keycloak, subject и issuer совпадают: openssl s_client -connect localhost:8443 -servername localhost </dev/null 2>/dev/null | openssl x509 -noout -subject -issuer -dates



Проверка готовности по HTTPS, ревизия 2: helm test kc-prod -n k8s-lab

